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INNOVATION. AUTOMATION. ANALYTICS

PROJECT ON

ANALYSIS ON PRICES OF USED CARS

About me

- **Pavani Munagala**
- **I Completed my graduation in Biotechnology Engineering from National Institute of Technology Raipur in 2024, where I worked on a exciting project named "Detection of Cancer Cells Using Linear Regression Models". That was a turning point for me, as it sparked a deep interest in the field of data driven research and AI.**
- **As I delved deeper into this project, I became fascinated by how machine learning could be applied to solve real -world biological problems. It was through this exposure that I realized my passion for Data Science a field that is not only booming today but also shaping the future across many industries.Motivated by this interest, I decided to pursue my career in Data Science.**
- **Linkedin:**www.linkedin.com/pavani-munagala
- **Github::**https://github.com/Pavani-Reddy111

- **Saideepu Balaraju**
- **I completed B.Tech in Computer Science Engineering from JNTUA 2025.**
- **During my final year, I worked on a project titled "Credit Card Fraud Detection using Predictive Modelling", which sparked my deep interest in the field of Data Science. Motivated by this interest, I began exploring real-world datasets, tools, and techniques used in data-driven decision-making. I am now eager to pursue a career in Data Science, where I can combine my technical skills and analytical mindset to solve meaningful problems.**
- **Linkedin:**linkedin.com/in/saideepu-balaraju
- **Github:**https://github.com/SAIDEEPUBALARAJU-2003

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Agenda

- Objective of the project
- Web Scraping
- Information about Collected Data

Exploratory Data Analysis:

a. Data Cleaning Steps
b. Data Manipulation Steps
c. Data Visualization
d. Analysis based on Features
e. Obsercations
f. Conclusion

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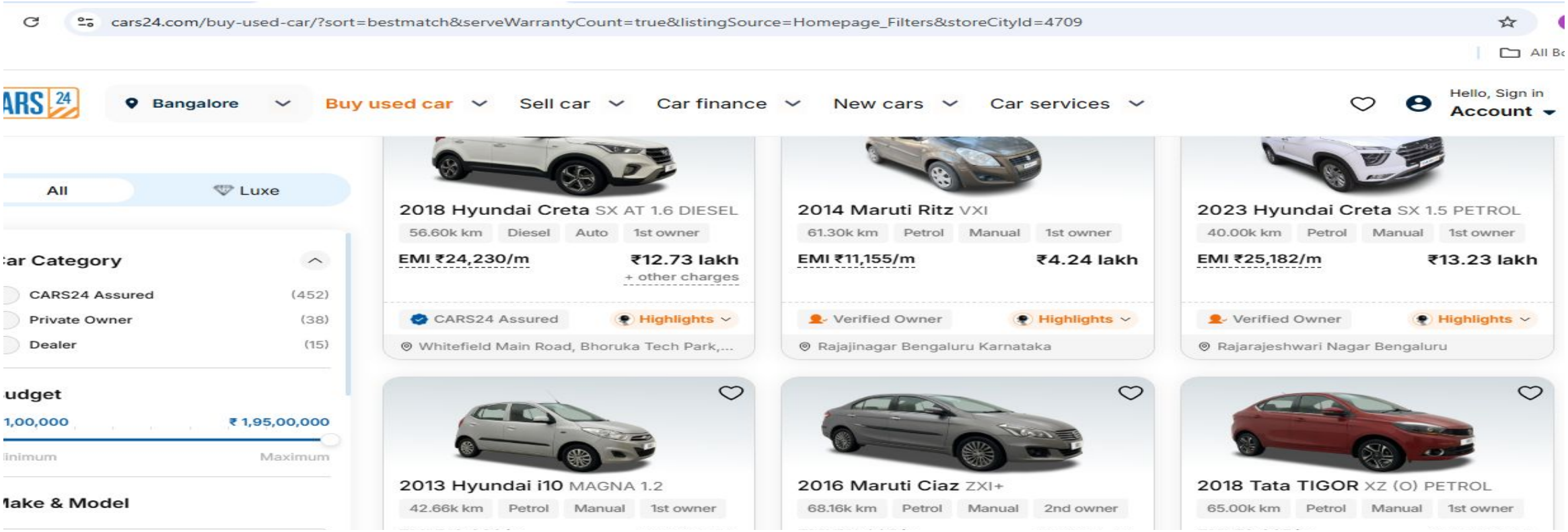
Objective Of Our Project

- Main goal of the project is to analyze historical car sales data to understand the features that impacts the price of a car.
- **Importance of this analysis:**
- This helps buyers who lacks technical knowledge about car depreciation, brand impact and market trends.
- This helps sellers to set competitive prices without undercutting their profits and also overestimating the price of car.
- This analysis helps buyers to find appropriate cars under budget.

Data Collection

- We collected data from Cars24 website.



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Data Collection Process

- Extracted the data using Requests and Beautiful Soup call with the help of Requests

```
[1]: import requests

[2]: url="https://www.cars24.com/buy-used-car/?sort=bestmatch&serveWarrantyCount=true&listingSource=Homepage_Filters&storeCityId=4709"

[3]: page = requests.get(url)

[7]: page

[7]: <Response [200]>

[9]: from bs4 import BeautifulSoup

[11]: soup = BeautifulSoup(page.text)

[13]: soup

[13]: <!DOCTYPE html>
<html lang="en"><head><meta charset="utf-8"/><link href="https://assets.cars24.com" rel="preconnect"/><link href="https://media.cars24.com" rel="precon
nect"/><link href="https://connect.facebook.net" rel="preconnect"/><link href="https://www.googletagmanager.com" rel="preconnect"/><link href="https://
www.google-analytics.com" rel="preconnect"/><link href="https://analytics.twitter.com" rel="preconnect"/><link href="https://www.facebook.com" rel="pre
connect"/><link href="https://www.google.com" rel="preconnect"/><link href="https://www.google.co.in" rel="preconnect"/><link href="https://googleads.
g.doubleclick.net" rel="preconnect"/><link href="https://t.co" rel="preconnect"/><link href="https://static.ads-twitter.com" rel="preconnect"/><link hr
ef="https://tracking.cars24.com" rel="preconnect"/><link href="https://gumi.criteo.com" rel="preconnect"/><link href="https://dynamic.criteo.com" rel
="preconnect"/><link href="https://tr.snapchat.com" rel="preconnect"/><link href="https://pixel.tapad.com" rel="preconnect"/><link href="https://sslwid
get.criteo.com" rel="preconnect"/><meta content="width=device-width, initial-scale=1, maximum-scale=5, user-scalable=yes" name="viewport"/><link data-p
recedence="next" href="https://assets.cars24.com/production/catalog-web-in/250417174129/_next/static/css/199514ba371a96df.css" rel="stylesheet"/><link
data-precedence="next" href="https://assets.cars24.com/production/catalog-web-in/250417174129/_next/static/css/fec8633f9ada2dd5.css" rel="stylesheet"/>
```

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Raw Data Overview

```
[288]: df.to_csv("Cars_dataset.csv")
       df
```

[288]:

	Kms_driven	Petrol_Diesel	Car_Type	Car_owner_Type	Price	Manufacture Year	Brand
0	59.27k km	Petrol	Manual	(2nd, owner)	₹2.73 lakh	2015	Hyundai i10
1	46.89k km	Petrol	Manual	(1st, owner)	₹2.62 lakh	2013	Maruti Wagon R 1.0
2	21.27k km	Petrol	Manual	(1st, owner)	₹2.37 lakh	2016	Renault Kwid
3	41.20k km	Petrol	Manual	(2nd, owner)	₹2.27 lakh	2016	Renault Kwid
4	39.74k km	Petrol	Manual	(1st, owner)	₹2.74 lakh	2016	Maruti Alto K10
...	...	...	...	...	...	...	...
869	17.54k km	Petrol	Auto	(1st, owner)	₹16.27 lakh	2021	MG HECTOR
870	69.95k km	Petrol	Auto	(1st, owner)	₹13.00L₹12.75 lakh	2022	KIA SELTOS
871	NaN	Diesel	Manual	(1st, owner)	₹13.17 lakh	2020	Tata Harrier
872	27.40k km	Petrol	Auto	(2nd, owner)	₹11.50L₹10.80 lakh	2019	MG HECTOR
873	51.33k km	Diesel	Manual	(2nd, owner)	₹13.06 lakh	2020	MG HECTOR PLUS

874 rows × 7 columns

Features Extracted

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 874 entries, 0 to 873
Data columns (total 7 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Kms_driven        721 non-null    object
 1   Petrol_Diesel     874 non-null    object
 2   Car_Type          874 non-null    object
 3   Car_owner_Type    828 non-null    object
 4   Price             848 non-null    object
 5   Manufacture Year  665 non-null    float64
 6   Brand             665 non-null    object
dtypes: float64(1), object(6)
memory usage: 54.6+ KB
```

.Total columns:7

.Total _rows:874

. Kms_driven,Fuel_Type,Car_type,

are extracted from cars24.

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Data Cleaning

- Replacing all the special Characters with space in the Car_owner_type and Kms_driven columns.

```
df["Car_owner_Type"] = df['Car_owner_Type'].str.replace(",","")
df["Car_owner_Type"] = df['Car_owner_Type'].str.replace(",","")
df["Car_owner_Type"] = df['Car_owner_Type'].str.replace("' '"," ")
df["Car_owner_Type"] = df['Car_owner_Type'].str.replace("'","  ")
df["Car_owner_Type"] = df['Car_owner_Type'].str.replace(")","")
df["Car_owner_Type"] = df['Car_owner_Type'].str.replace("(","")
df["Kms_driven"] = df['Kms_driven'].str.replace("m","")
```

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Data Cleaning

- Replacing valid prices using regular expressions in Price column.
- Converting Kms driven values to numeric by removing k and multiplying it with 1000.

```
df["Price"] = df["Price"].apply(lambda x : re.findall(r"₹\d+\.\d+\slakh",str(x))[0] if re.findall(r"₹\d+\.\d+\slakh",str(x)) else np.nan)

df["Price"] = df['Price'].str.replace("₹","")

df["Price"] = df['Price'].str.replace("lakh","")

def kms_driven(i):
    if i==np.nan:
        return i
    elif 'k' in str(i):
        a=float(i.replace('k',''))*1000
        return a
    else:
        return i

df["Kms_driven"]=df["Kms_driven"].apply(kms_driven)
```

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Data Cleaning:

- Dropping the null values from brand column.
- Dropping the null values from Kms_driven and price column.
- Very few null values are present in both.

```
df.dropna(subset=["Brand"],inplace=True)
```

```
df[(df["Kms_driven"].isna() & df["Price(lakh)"].isna())].index
```

```
Index([119, 391, 483, 489, 599, 673, 699, 711, 713, 720, 744, 768, 769, 773,
       782, 791, 799, 816],
      dtype='int64')
```

```
df.drop([119, 391, 483, 489, 599, 673, 699, 711, 713, 720, 744, 768, 769, 773,782, 791, 799, 816],inplace=True)
```

```
df[ df["Price(lakh)"].isna()].index
```

```
Index([381, 493, 553, 662, 679, 752, 790], dtype='int64')
```

```
df.drop([381, 493, 553, 662, 679, 752, 790],inplace=True)
```

Data Cleaning

- Counting the values from Car_owner_type column and filling the null values with maximun count i.e. 1st owner.

```
df["Car_owner_Type"].value_counts()

Car_owner_Type
1st  owner     487
2nd  owner     146
Name: count, dtype: int64

df["Car_owner_Type"].fillna(' 1st  owner ',inplace=True)
```

- Using group by imputation on Car_owner_type and Kms_driven column with agg func mean to fill null values in kms_driven column.

```
df.groupby(by="Car_owner_Type")["Kms_driven"].agg("mean")

Car_owner_Type
1st  owner     48657.326733
2nd  owner     58586.000000
Name: Kms_driven, dtype: float64

df.loc[(df['Kms_driven'].isna()) & (df['Car_owner_Type'] == ' 1st  owner '), 'Kms_driven'] = 48842.0
df.loc[(df['Kms_driven'].isna()) & (df['Car_owner_Type'] == ' 2nd  owner '), 'Kms_driven'] = 58958.0
```

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Data Cleaning:

- Finding the car age using current year and maufacture year and creating a new column named “Car_Age”.

```
from datetime import datetime
current_year=datetime.now().year
```

```
df["Car_age"]=current_year-df["Manufacture Year"]
```

```
df.rename(columns={"Petrol_Diesel": "Fuel_Type"},inplace=True)
```

```
df.rename(columns={"Price": "Price(lakh)"},inplace=True)
```

- Splitting a single column which consist of brand and model into two separate columns named “Brand” and “Model”.

```
df["brand"]=df["Brand"].str.split(" ",expand=True)[0]
```

```
df["Model"]=df["Brand"].str.split(" ").apply(lambda x : " ".join(x[1:]))
```

```
df.drop(columns=["Brand"],inplace=True)
```

Data Cleaning:

- Converting the data types of Price, Kms_driven and Manufacture_Year columns.

```
df["Price(lakh)"]=df["Price(lakh)"].astype(float)
df["Kms_driven"]=df["Kms_driven"].astype(float)
df["Manufacture Year"]=df["Manufacture Year"].astype(int)
```

. Resetting the index.

```
df.reset_index(inplace=True)
```

Data After Cleaning

- Final data after cleaning where I left with 640 rows and 8 Columns.

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 874 entries, 0 to 873
Data columns (total 7 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Kms_driven        721 non-null    object
 1   Petrol_Diesel     874 non-null    object
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 4   Price             848 non-null    object
 5   Manufacture Year  665 non-null    float64
 6   Brand             665 non-null    object
dtypes: float64(1), object(6)
memory usage: 54.6+ KB
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 640 entries, 0 to 639
Data columns (total 9 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Kms_driven        640 non-null    float64
 1   Fuel_Type         640 non-null    object
 2   Car_Type          640 non-null    object
 3   Car_owner_Type    640 non-null    object
 4   Price(lakh)       640 non-null    float64
 5   Manufacture Year  640 non-null    int32
 6   Car_age           640 non-null    float64
 7   brand             640 non-null    object
 8   Model             640 non-null    object
dtypes: float64(3), int32(1), object(5)
memory usage: 42.6+ KB
```

Count of Brands and Models in the Dataset

```
df["brand"].nunique()
```

```
20
```

```
df['Model'].count()
```

```
640
```

```
df.groupby(by="brand")["Model"].agg("count").sort_values(ascending=False)
```

```
brand
Maruti        194
Hyundai       134
Tata           78
Renault        42
Mahindra       38
Honda          33
Volkswagen     21
Nissan         18
KIA            17
Skoda          14
Toyota         13
MG             11
Ford           10
Jeep            7
Datsun          4
Chevrolet       2
BMW             1
Mercedes        1
Fiat            1
Audi            1
Name: Model, dtype: int64
```

- 20 different brands are present in our dataset.
- 640 models from the brands are present in our dataset.

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Price Range of Cars in the dataset

- Maximum price of the car is 39 lakhs.
- Minimum price of the car is 82 thousand.
- Average price of the cars are around 6 lakhs.

```
df["Price(lakh)"].max()
```

```
39.9
```

```
df["Price(lakh)"].min()*100000
```

```
82000.0
```

```
df["Price(lakh)"].describe()
```

```
count    640.000000
mean       6.039250
std        3.936821
min        0.820000
25%        3.515000
50%        5.295000
75%        7.200000
max       39.900000
Name: Price(lakh), dtype: float64
```

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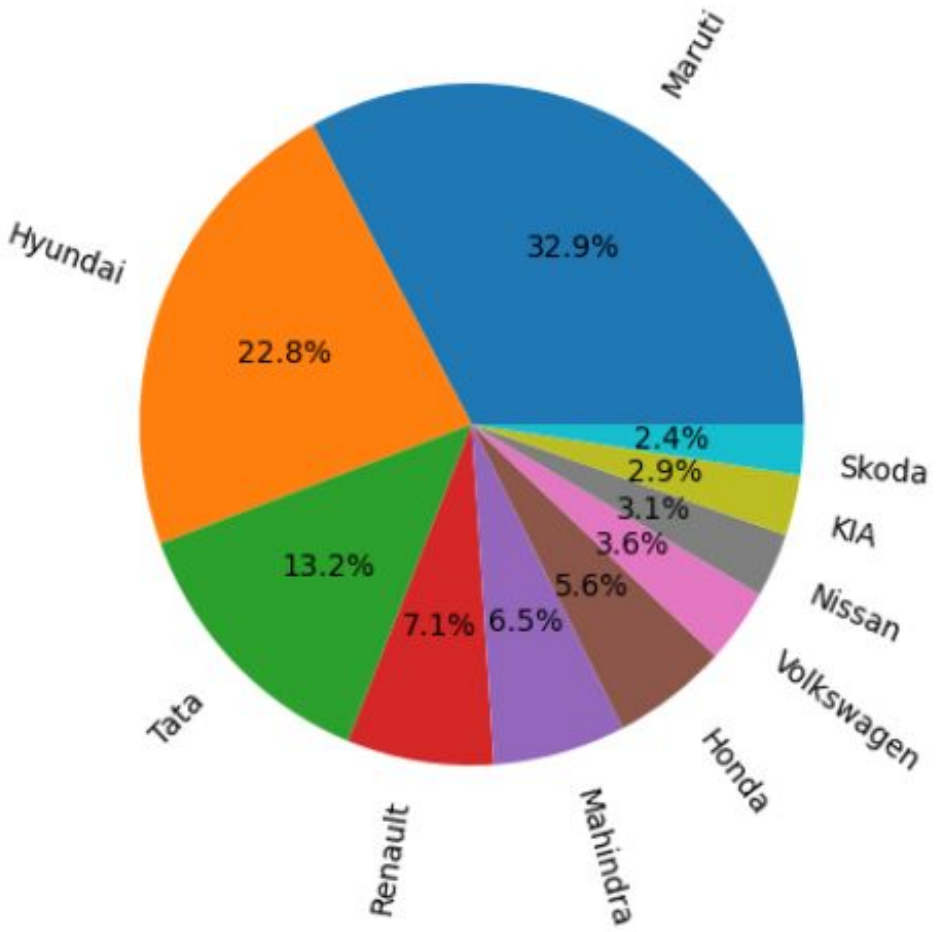
Top 10 Brands Percentage distribution



Major Cars:

32.9% cars are from Maruti.

22.8% cars are from Hyundai.

13.2% cars are from Tata.

Minor Cars:

3.1% cars are from Nissan.

2.9% cars from KIA.

2.4% are from Skoda.

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Maximum and Minimum Car prices by brand

Maximum priced cars are Mercedes, Benz,Audi

Minimum priced cars are Maruti,Hyundai, Tata

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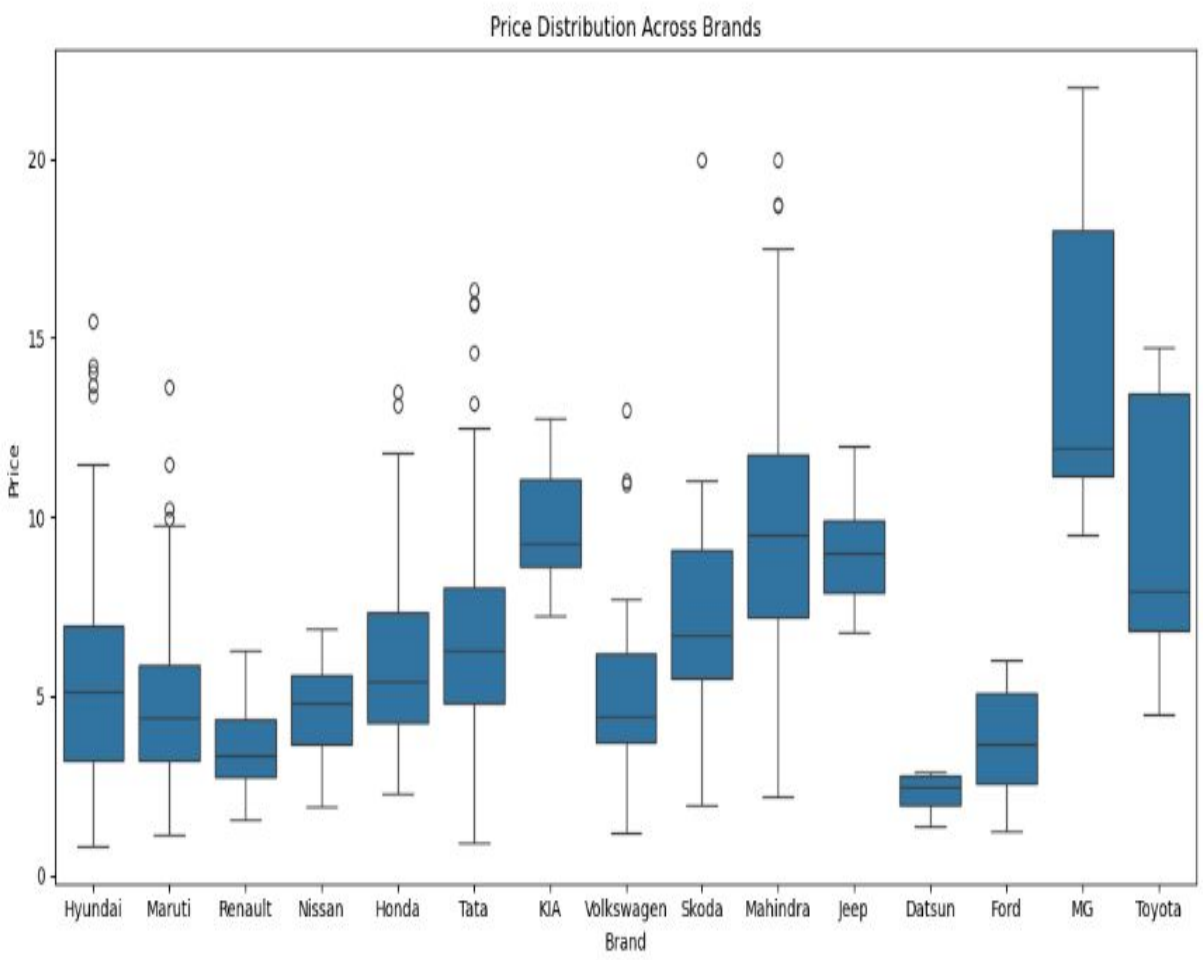
Price Distribution Across Brands



Hyundai, Maruti, Tata manufactures both budget and luxury models .

Mahindra mostly manufactures premium and high range cars.

MG manufactures only high range cars

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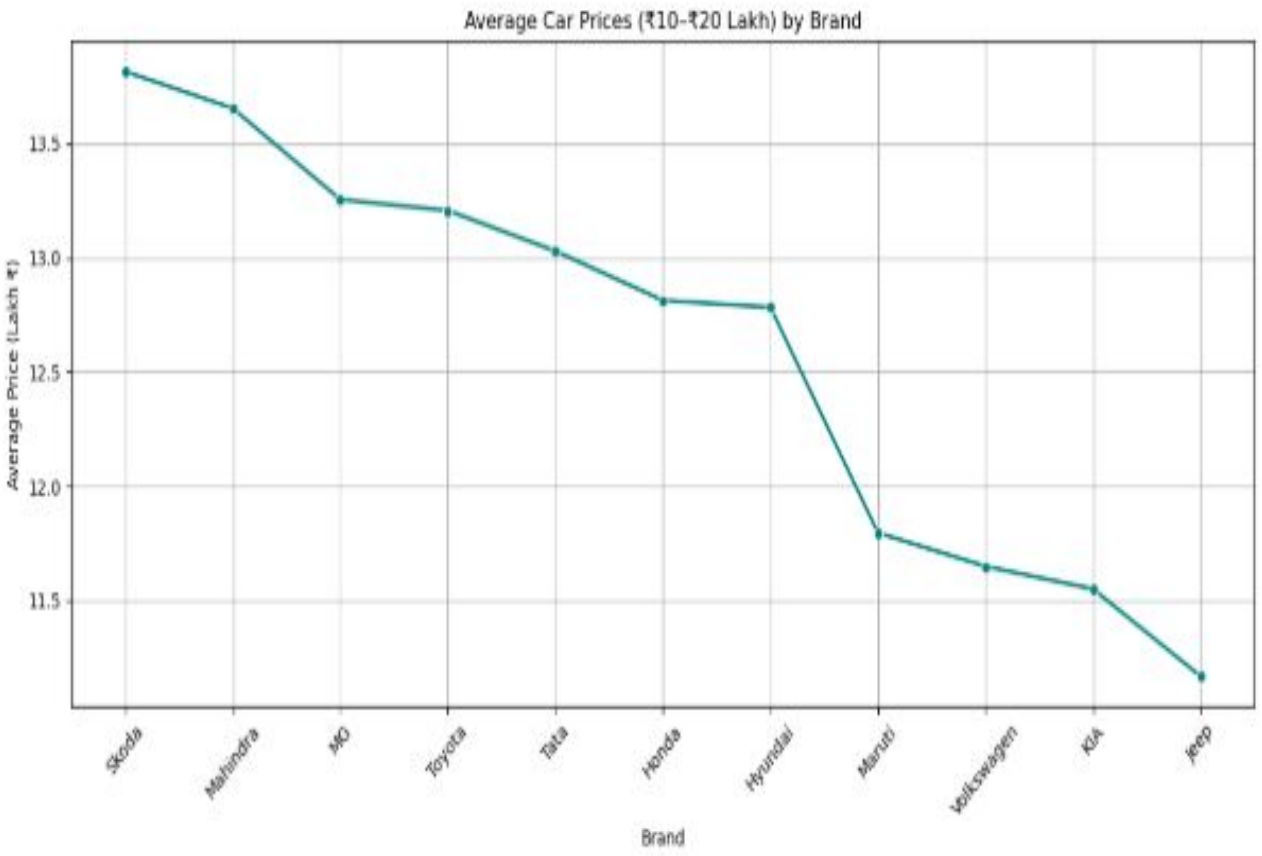
Average Car prices(10-20 lakhs) by brand



11 Brands falls in the average price range 10 to 20 lakhs

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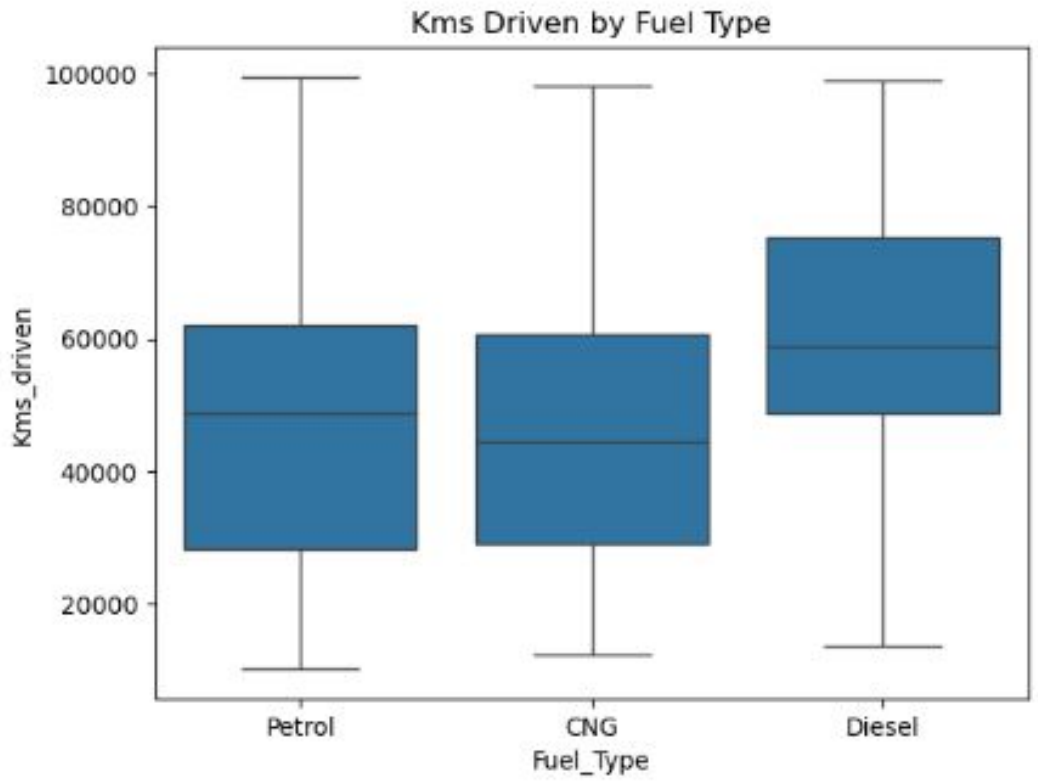
Kilometers Driven by Fuel Type



Graph shows that Diesel cars gives more mileage than petrol,CNG Cars

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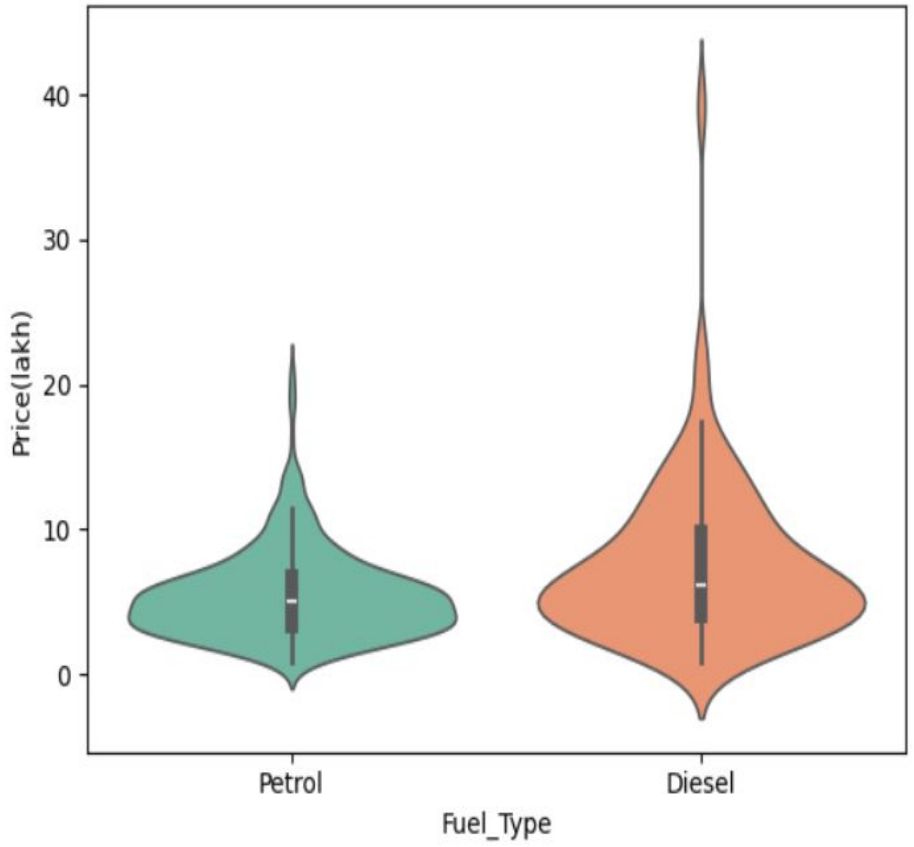
Price Variation Across Different Fuel Types



- Most Petrol cars price falls in the between 0 to 15.5 lakh .
- Most Diesel cars price fall in the between 0 to 25 lakh .
- 50% of petrol cars are around 7 lakh .
- 50% of diesel cars are around 8 lakh .

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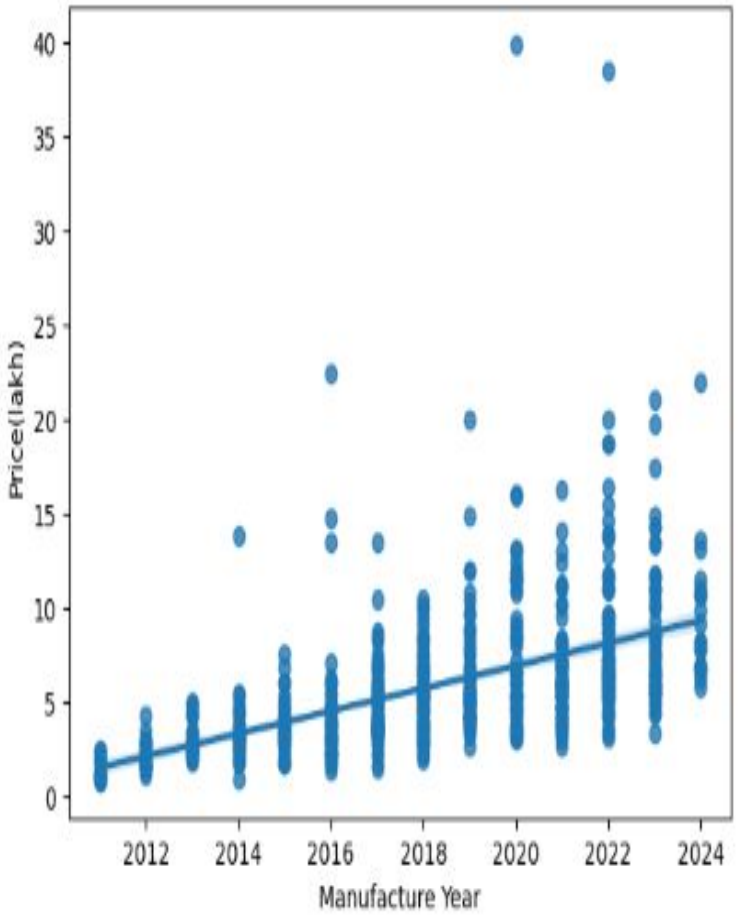
Impact of Manufacturing Year on Car Price



From this graph we can analyze that as the increase in Manufacturing Year car price is also increases

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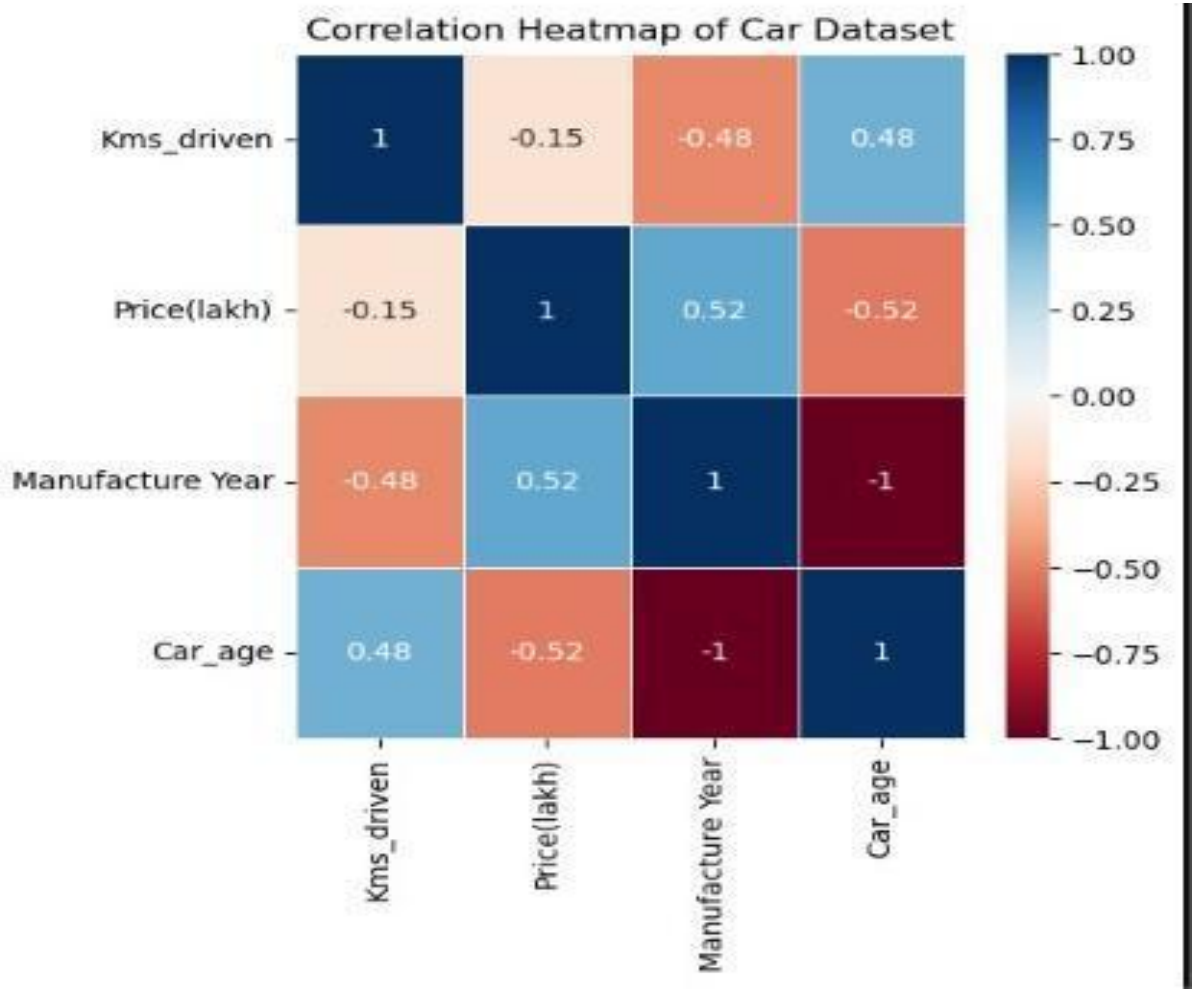
Correlation Heatmap of Car Dataset



. Negative relationship between Price and Kms driven.

. Positive relationship between Manufacturing Year and Price.

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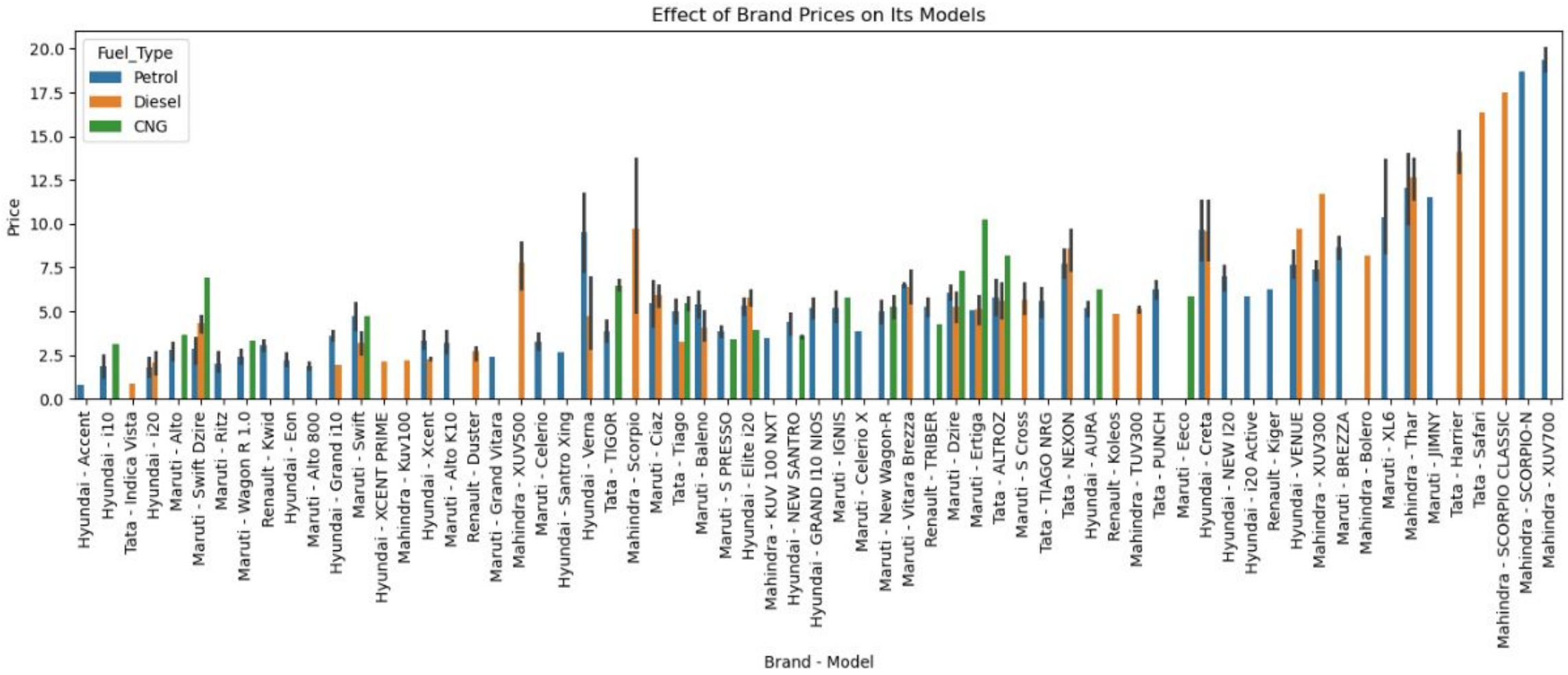
Effect of Brand Prices On Its Models



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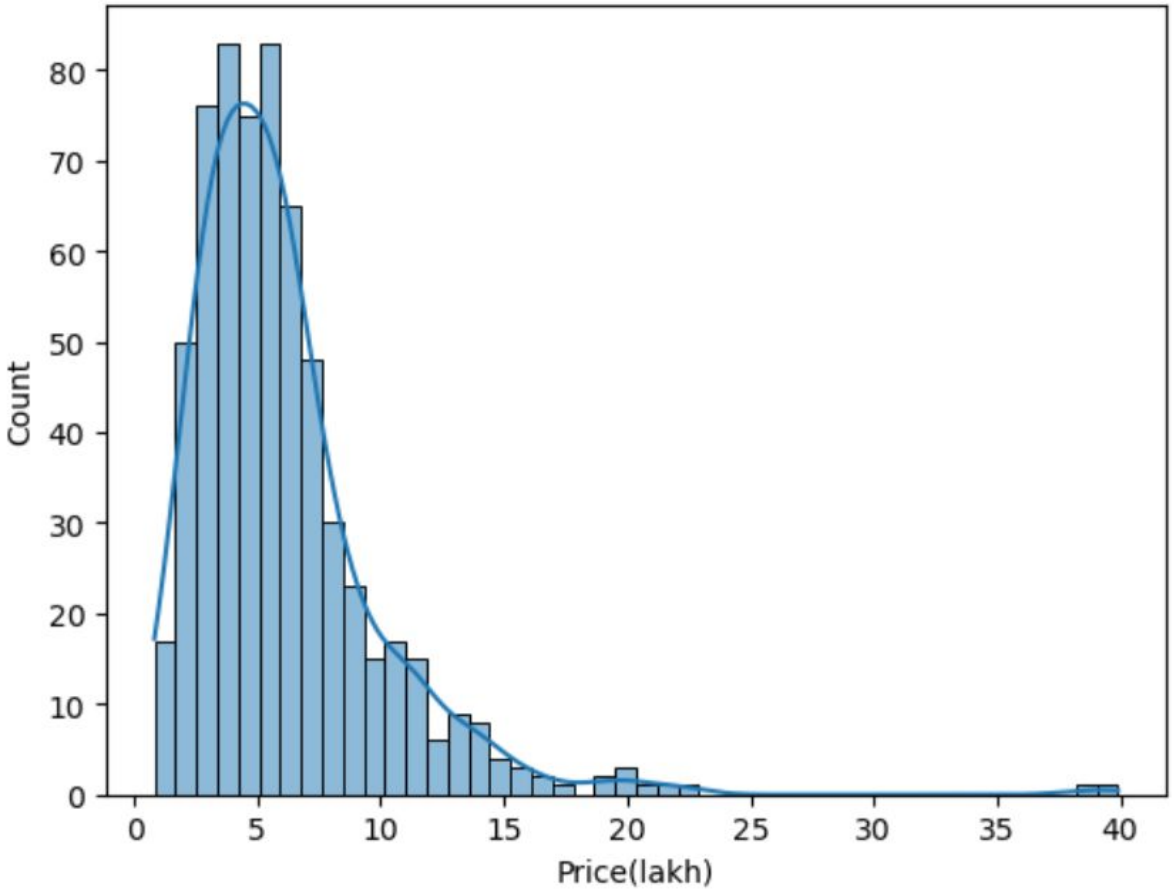
Price Range Analysis of Used Cars



There are more cars in price between 2.5 lakhs to 6.5 lakhs

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Observations

After detailed analysis of the project we got the following insights:

- Price decreases significantly as the car get older
- 1st owner cars have high resale value than 2nd owner cars
- Brands like BMW, Mercedes, Audi retain better value
- Petrol cars are often cheaper than Diesel cars
- Major price range of cars are between 2.5 to 6.5 lakhs
- New model cars are of high price
- Diesel cars gives more mileage

Conclusion:

- Car price is most influenced by age,mileage,brand,fuel type
- High brand cars are of high price and have high resale value
- There are less CNG cars available in the market
- Major budget cars are from Maruti,Hyundai
- Diesel cars are of high price range compare to petrol cars

THANK YOU

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